Kefan Wu

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EDUCATION

Olin College of Engineering
Needham, MA · Expected May 2028

B.S. Mechanical Engineering — GPA 3.9/4.0

Coursework: Mechanics of Solids & Structures, Systems Analysis: Dynamics, Thermal-Fluid Systems, Probabilistic Modeling

EXPERIENCE

Olin Electric Motorsports *(Formula SAE Electric)*
Needham, MA

Student team: built an 85 kW electric racecar from scratch; 0–60 mph in 3.5 s.

Mechanical Lead — May 2026 – Present

- Direct mechanical design, integration, and fabrication of the Mk.8 FSAE electric racecar
- Manage **4 mechanical subteams**: suspension & chassis, aerodynamics, drivetrain, accumulator
- Set milestones and build schedules toward a **22 km, sub-30-minute** endurance design target

Cockpit Lead — Jun 2025 – May 2026

- Developed a track-based brake thermal model; cut brake rotor mass **23%** at a **2.455 FEA factor of safety**
- Engineered the steering column around matched **27.5° dual U-joints** providing linear steering; lightweighted steering bearing cages, saving **0.9 kg** against last year's design
- Led the cockpit subteam: machined parts on lathe, mill, and waterjet; TIG-welded chassis tabs; completed a carbon fiber seat

Drivetrain & Cockpit Engineer — Sep 2024 – Jun 2025

- Designed and fabricated the **5052-H32 aluminum** Mk.7 driver seat and FEA-validated motor-controller mounts

Research Assistant, Olin Vine Robotics Lab *(full-time summer research)*
May 2026 – Aug 2026

- Built a vine-everting robot; reinforced vessel designed for **5 psi** via FEA, 3.3x its 1.5 psi max working pressure
- Determined material properties with **30 Instron tests** across TPU-coated fabric and LDPE (ASTM D882 standard)
- Designed a testing rig with minimized buckling to validate a cross-section strain model against the data

Mechanical Engineer, Project AURA *(class project)*
Oct 2025 – Dec 2025

- Designed front swerve-drive modules for an autonomous luggage robot: **200 lb target**, **300 lb tested**. NEMA 23 steppers via 18:80 chain steering reduction, 24 V MY1016Z DC drive motors via a 9:16 sprocket ratio.
- Diagnosed and fixed chain skip, backlash, and shaft misalignment in durability, impact, and load tests
- Waterjet-cut 0.25 in A36 steel wheel housings and motor mounts on an OMAX; TIG-welded the assemblies

SELECTED PROJECTS

- **Javelin VTOL drone** (in progress) — tail-sitter, no control surfaces, **300 km/h** design target; differential-thrust attitude control, avionics integrated; seeking FAA clearance to fly past the 100 mph UAS limit
- **Agent-based CFD** — automated Ansys Fluent via PyFluent with AI agents; drove Javelin VTOL aerodynamics design inspiration; documented **11 failure modes** with recovery procedures
- **LiDAR 3D scanner** — two-axis ESP32 gantry (NEMA 17, DRV8825) captured **2,206 points**; a 14-point calibration cut range error **below 3%**, plus copper-foil EMI shielding

SKILLS

CAD	SolidWorks (topology studies), AutoCAD, design for manufacturing
Analysis & Simulation	MATLAB, FEA (SolidWorks Simulation), CFD (Ansys Fluent, PyFluent), analytical hand calculations
Manufacturing & Test	CNC mill, lathe, CNC waterjet, plasma cutter, TIG welding, carbon fiber layup, sheet metal, SLA/FDM 3D printing, laser cutting, rapid prototyping, Instron testing
Software & Electronics	Arduino, ArduPilot, stepper control, PID control, agent-based simulation